AI-Based Smart Resume Screening System

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ABSTRACT

The growing number of job applications has made manual resume screening time-consuming, inefficient, and prone to bias. HR professionals face information overload, increasing the risk of missing qualified candidates.

Traditional ATS systems rely on exact keyword matching, lacking semantic understanding and often rejecting candidates who use synonymous terms. These systems are also vulnerable to keyword stuffing.

Additionally, many AI-based screening tools lack transparency, acting as “black boxes” that provide scores without clear explanations, raising fairness and trust concerns.
These issueshighlighttheneedfor a smarter, transparent, andsemantically intelligent resume screening system.

EXISTING SYSTEM

- Current resume screening systems primarily include manual evaluation by recruiters, traditional Applicant Tracking Systems (ATS), and basic NLP-based filtering tools. These systems assist in organizing and filtering large volumes of applications by using keyword matching, rule-based criteria, and predefined filters. While they provide basic support in handling recruitment workflows, they rely heavily on static logic and lack the ability to understand the contextual meaning of candidate profiles.
- Most existing systems evaluate resumes based on exact keyword matches and manually assigned importance to specific skills or qualifications. They do not effectively recognize semantic similarity between different terms or phrases, which leads to the rejection of qualified candidates who use alternative wording. Additionally, these systems do not consider deeper relationships between skills, experience, and job requirements, resulting in limited accuracy in candidate shortlisting.
- Furthermore, conventional resume screening approaches lack transparency and adaptability. Many AI-based systems function as black boxes, providing ranking scores without clear explanations. They do not incorporate explainable mechanisms, bias control, or dynamic learning based on recruiter feedback. As a result, these systems fail to ensure fairness, accountability, and efficient decision-making in modern recruitment processes.

PROPOSED SYSTEM

The proposed system is an Explainable AI-based resume screening solution that uses LLaMA to understand resumes and job descriptions based on meaning rather than keywords. It generates semantic embeddings, ranks candidates using similarity scores, and provides clear explanations for each decision. This approach improves efficiency, accuracy, and fairness in the recruitment process.

Key Features of the Proposed System

1. Automated Resume Processing
   - Supports resumes in PDF/DOC formats
   - Extracts structured data such as:
     - Skills
     - Experience
     - Education
     - Certifications
2. Semantic Matching using LLaMA
   - Converts resumes and job descriptions into embeddings
   - Understands contextual meaning instead of keywords
   - Handles synonyms and varied terminology

3. Similarity-Based Candidate Ranking
- Uses cosine similarity for comparison
- Generates relevance scores
- Ranks candidates based on matching level

4. Threshold-Based Filtering
- Filters candidates using similarity cutoff
- Removes irrelevant profiles automatically
- Improves precision of shortlisting

5. Explainable AI (XAI) Layer
- Provides reasons for candidate selection
- Highlights:
  - Matched skills
  - Relevant experience
  - Builds trust and transparency

6. Reasoning & Intelligence Layer
- Evaluates:
  - Skill relevance
  - Experience alignment
  - Provides deeper understanding beyond surface-level matching

7. Bias Reduction Mechanism

- Focuses on skill-based evaluation
- Avoids dependency on personal attributes
- Supports fair hiring practices

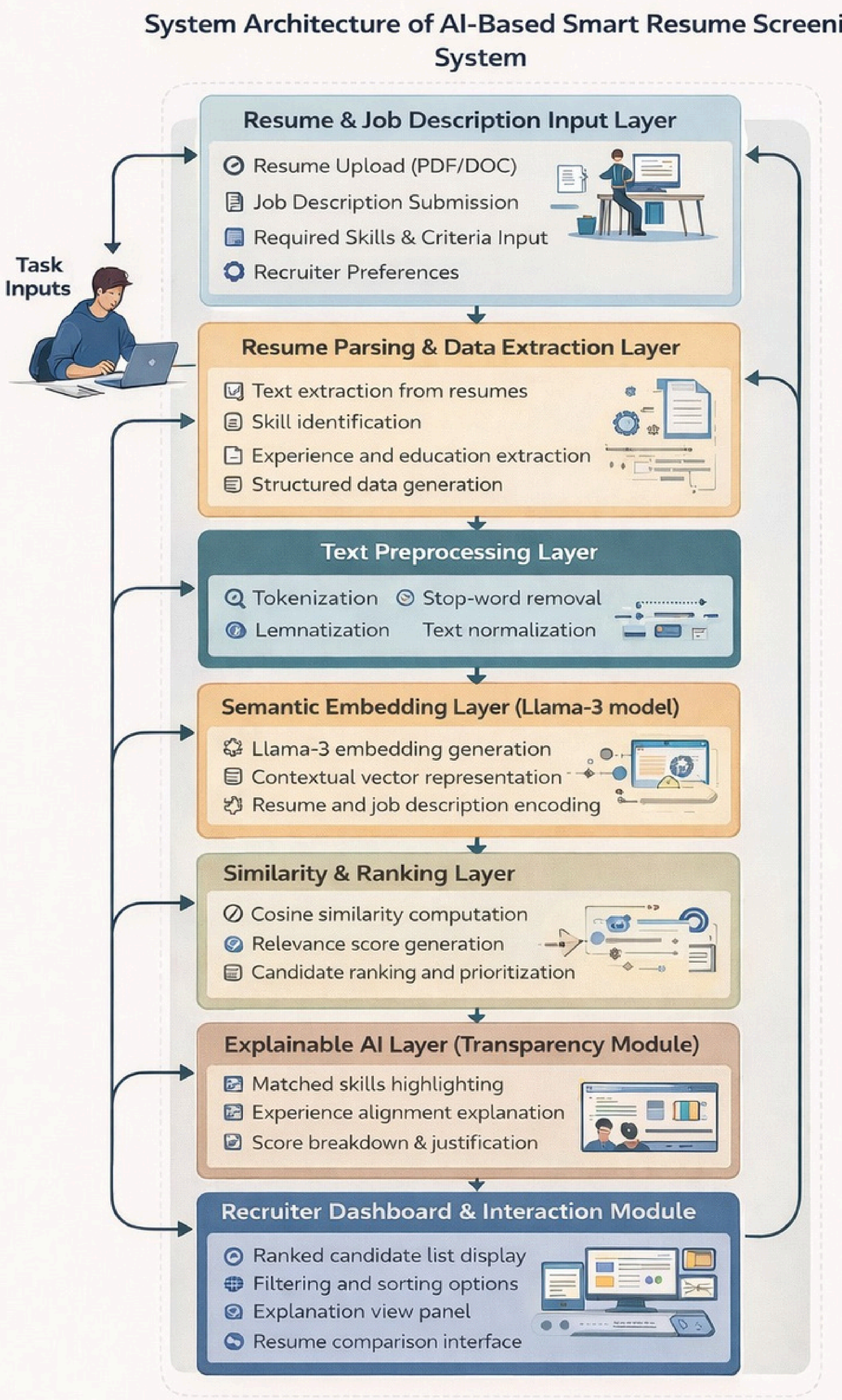
8. Continuous Improvement (Feedback Learning)

- Can incorporate recruiter feedback
- Improves ranking accuracy over time
- Adapts to changing job requirements

9. Interactive Dashboard

- Upload job descriptions and resumes
- View ranked candidates
- Analyze explanations
- Manage recruitment workflow efficiently

System Architecture



IMPLEMENTATION

1.Frontend (React.js + TypeScript)
- Resume upload & result dashboard
- Candidate ranking & filtering
- Explanation view

2.AI Integration (LLaMA / Sentence-Transformers)
- Semantic embedding generation
- Context-based matching

3.NLP & Resume Processing
- Text extraction & preprocessing
- Skill, experience extraction

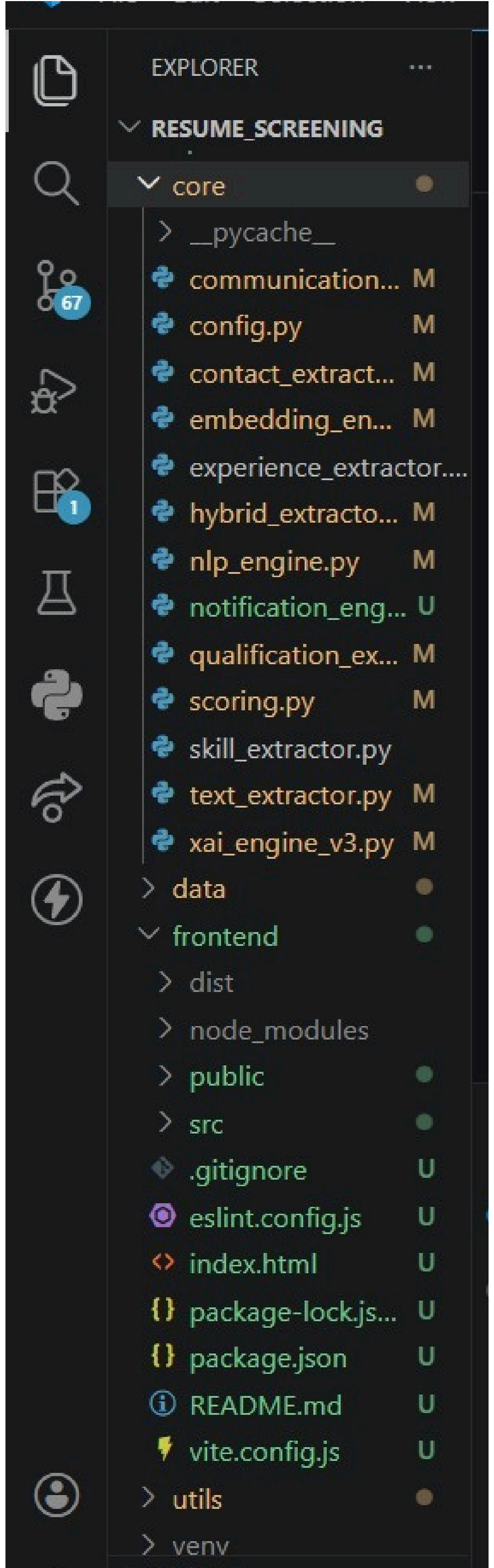
4.Backend (Python – Flask/FastAPI)
- API handling & data processing
- Integration of all modules

5.Data Storage
- Resumes, scores, preferences

6.Notification Module
- Email alerts to candidates

7. Similarity & Ranking
- Cosine similarity
- Candidate scoring & ranking

8.Explainable AI (XAI)
- Skill matching highlights
- Score justification


EXPLORER
RESUME_SCREENING
core
__pycache__
communication... M
config.py M
contact_extract... M
embedding_en... M
experience_extractor....
hybrid_extracto... M
nlp_engine.py M
notification_eng... U
qualification_ex... M
scoring.py M
skill_extractor.py
text_extractor.py M
xai_engine_v3.py M
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dist
node_modules
public
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.gitignore U
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README.md U
vite.config.js U
utils
venv
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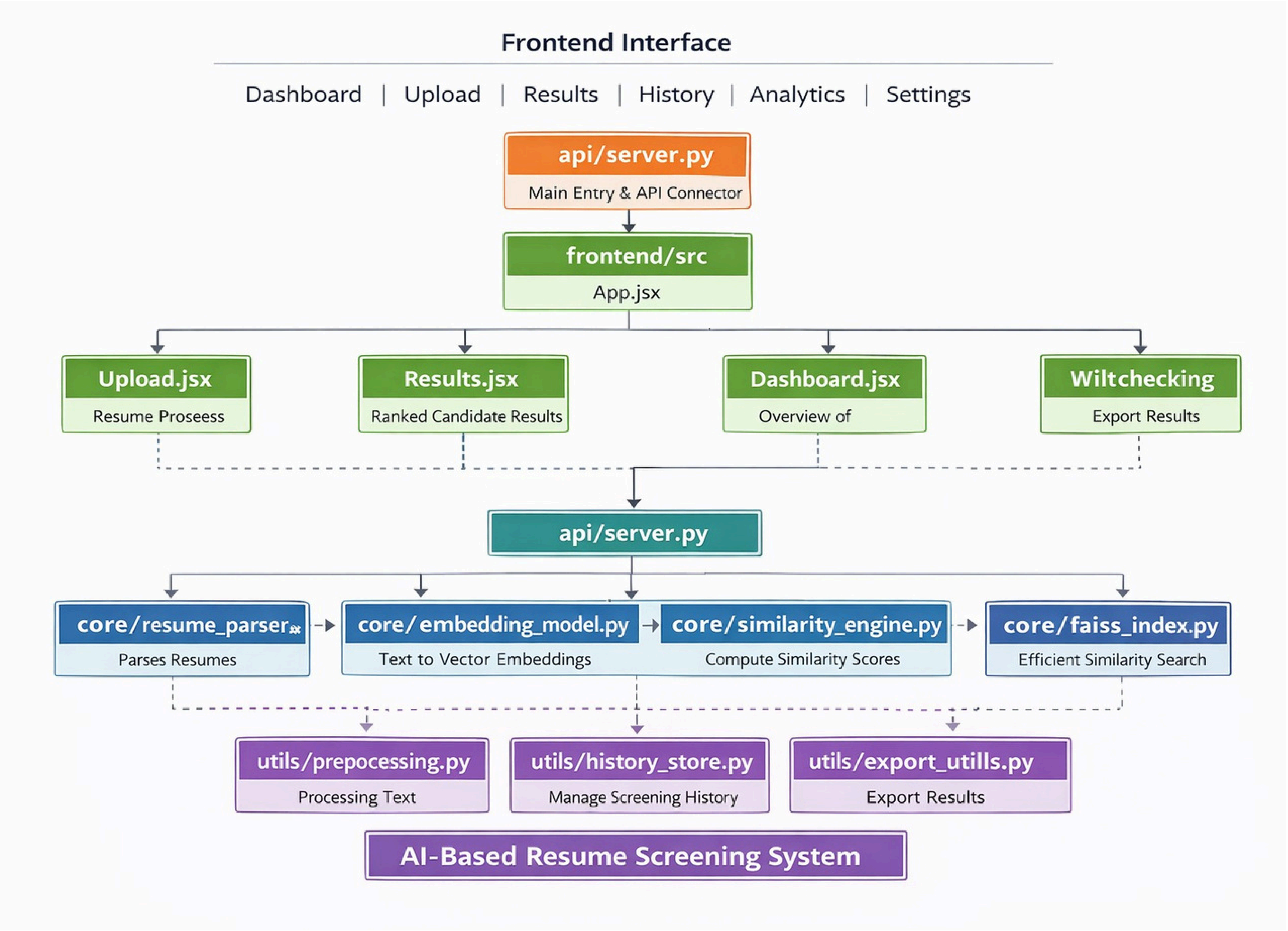
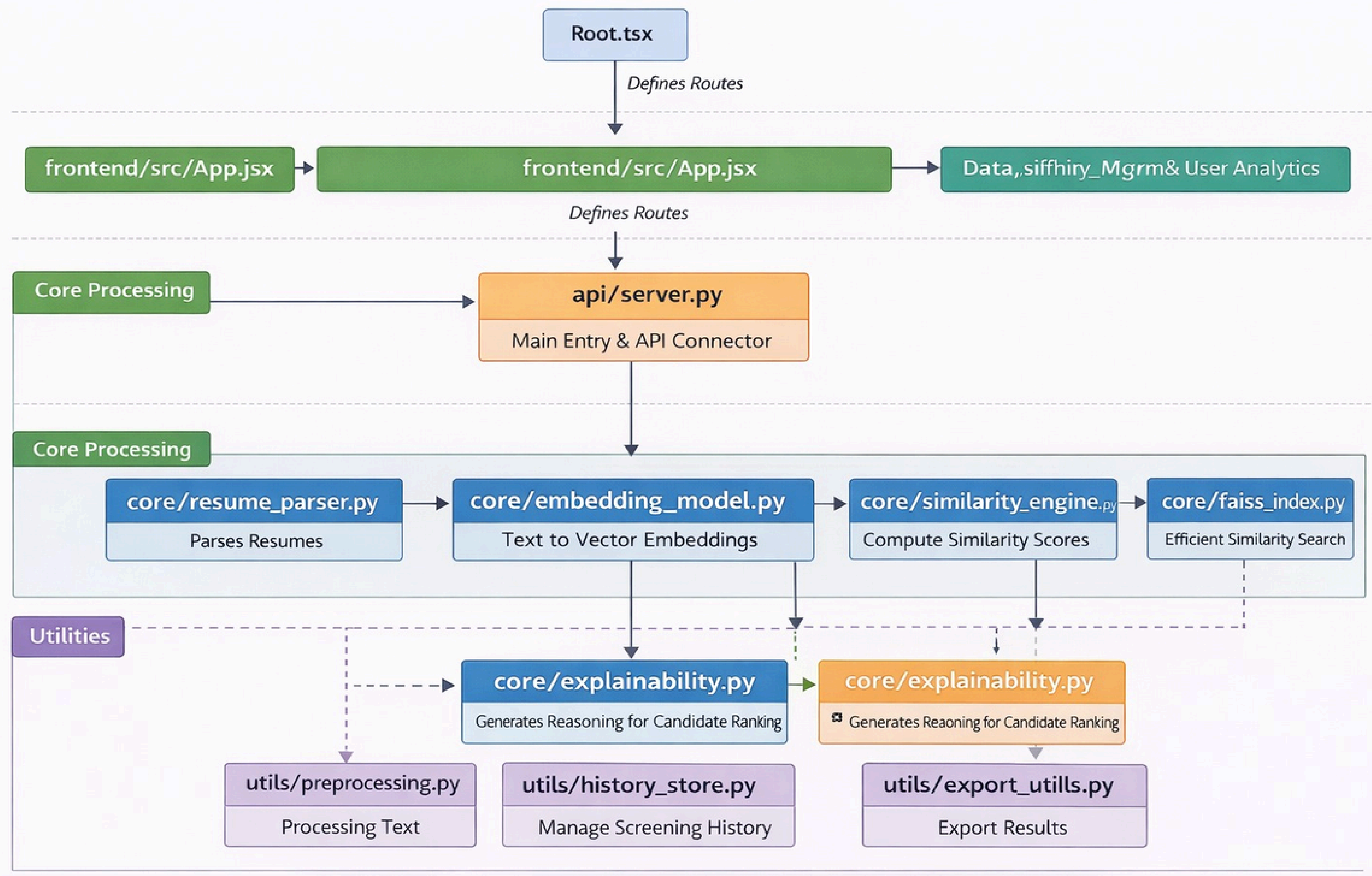
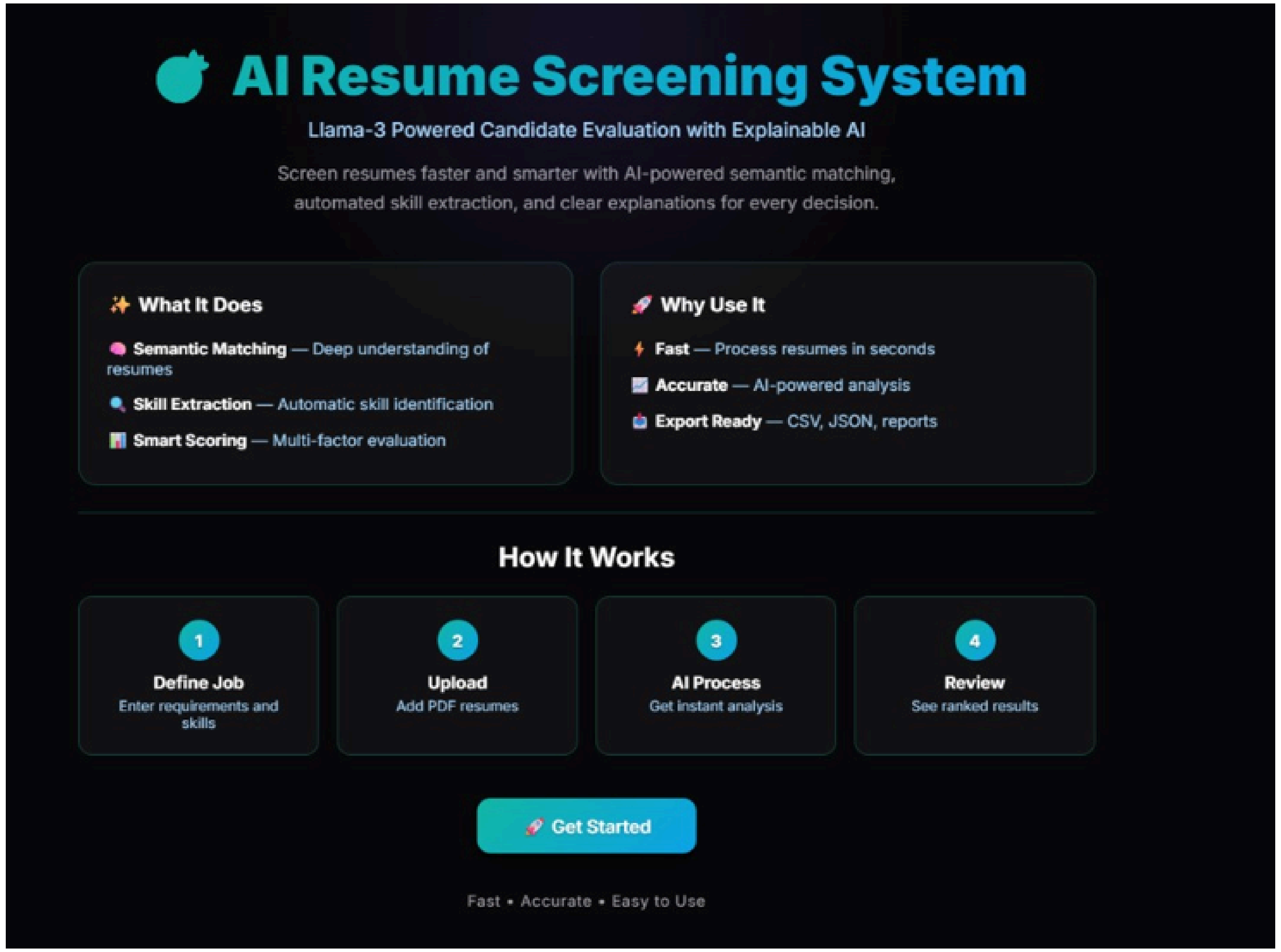


Layered Execution Flow



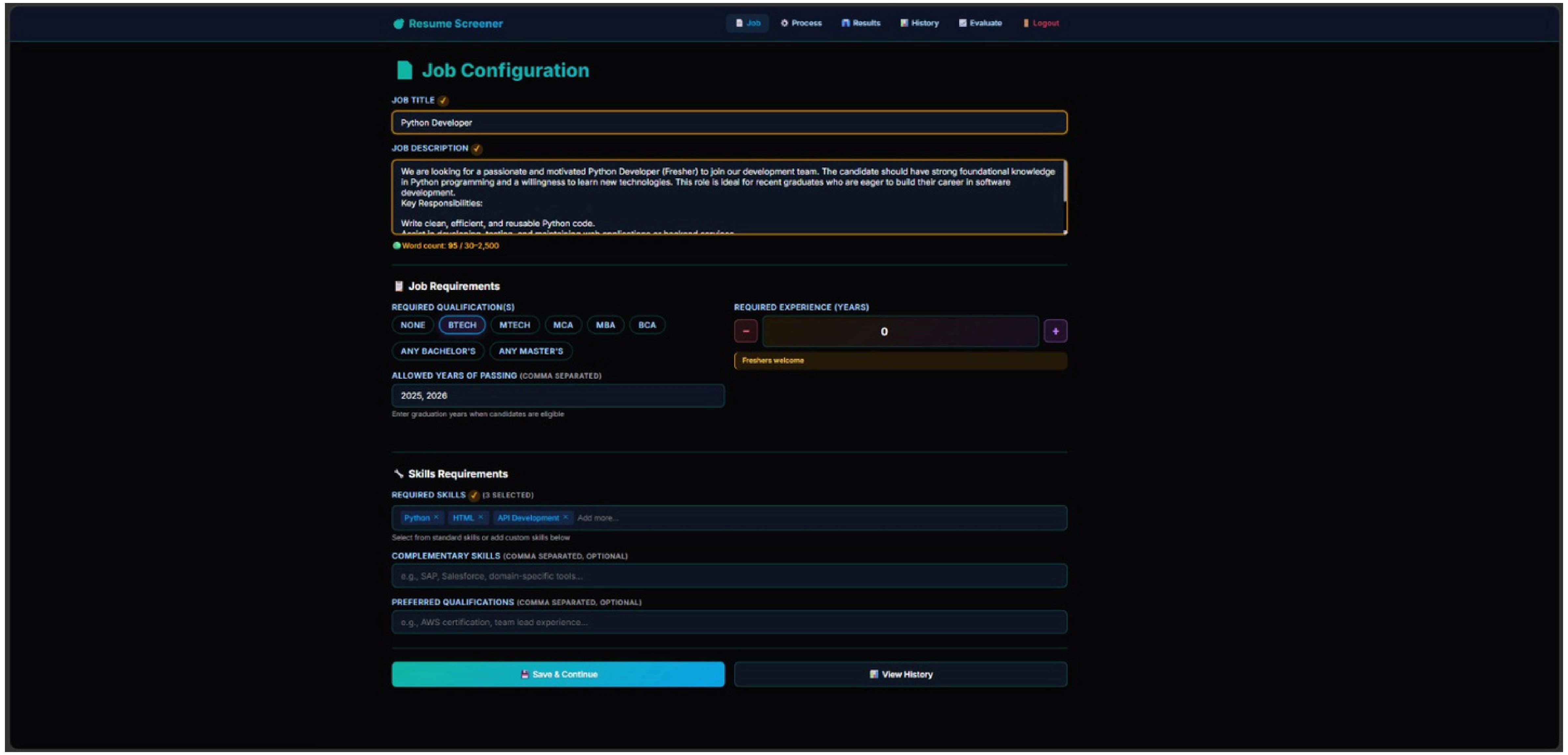
Detailed File Interaction Flow in AI-Based Resume Screening System

Results

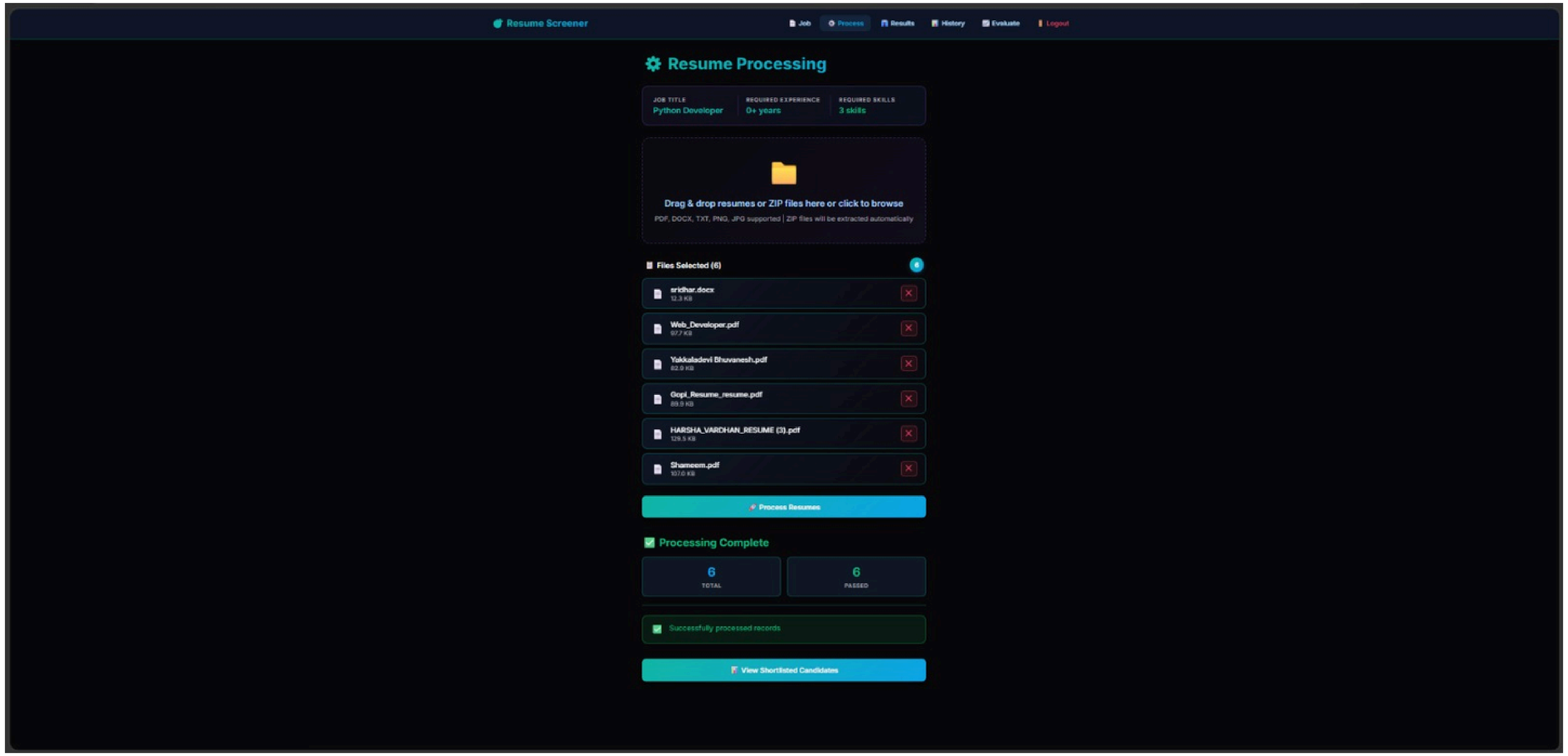


The home page provides an intuitive interface to define job requirements, upload resumes, and access AI-based screening with semantic matching and explainable results.

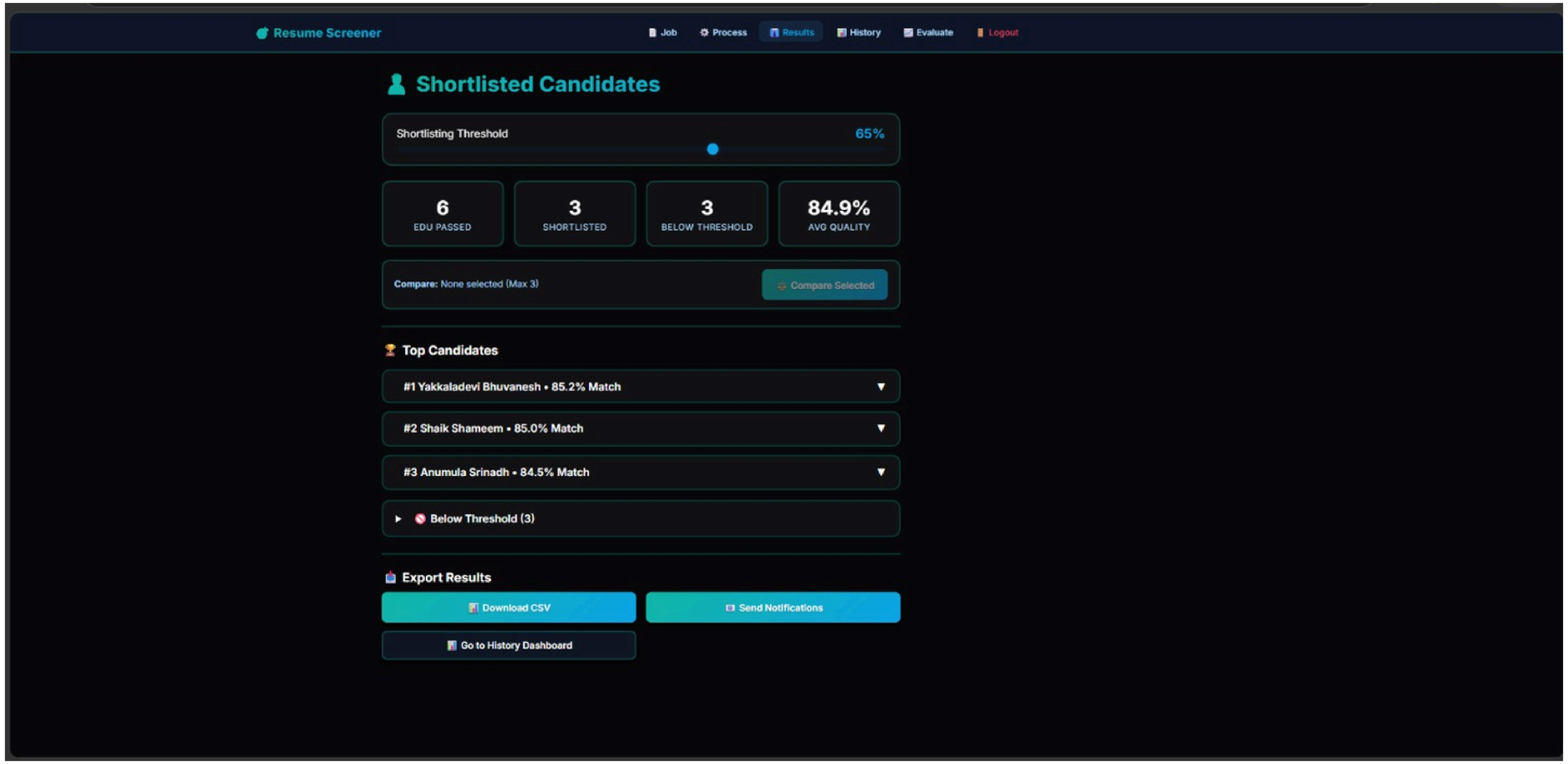
This page allows recruiters to define job roles, skills, qualifications, and experience, enabling accurate semantic matching and better candidate ranking.



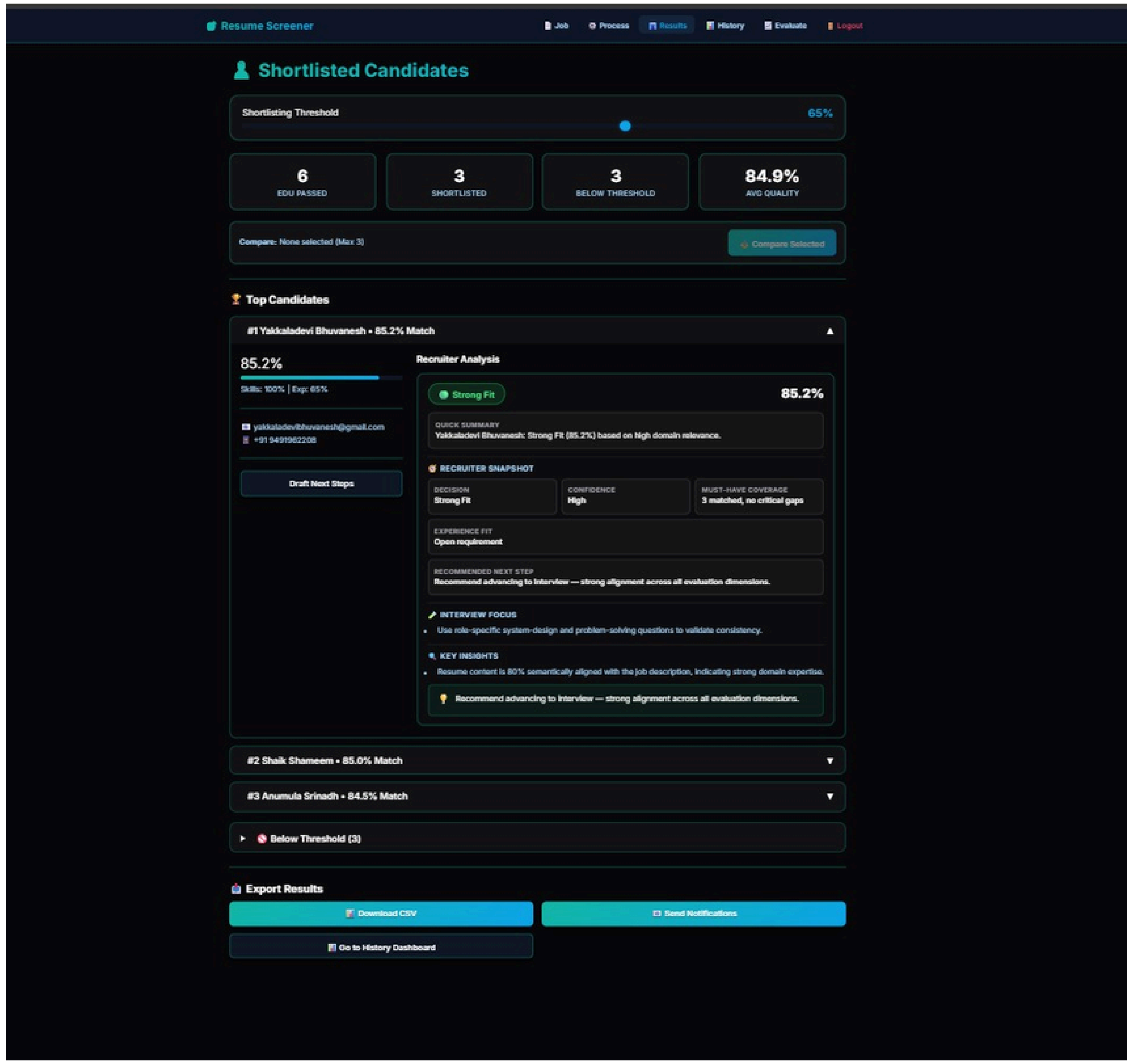
This page enables resume upload and automated processing using AI, generating embeddings, evaluating candidates, and displaying shortlisted results.



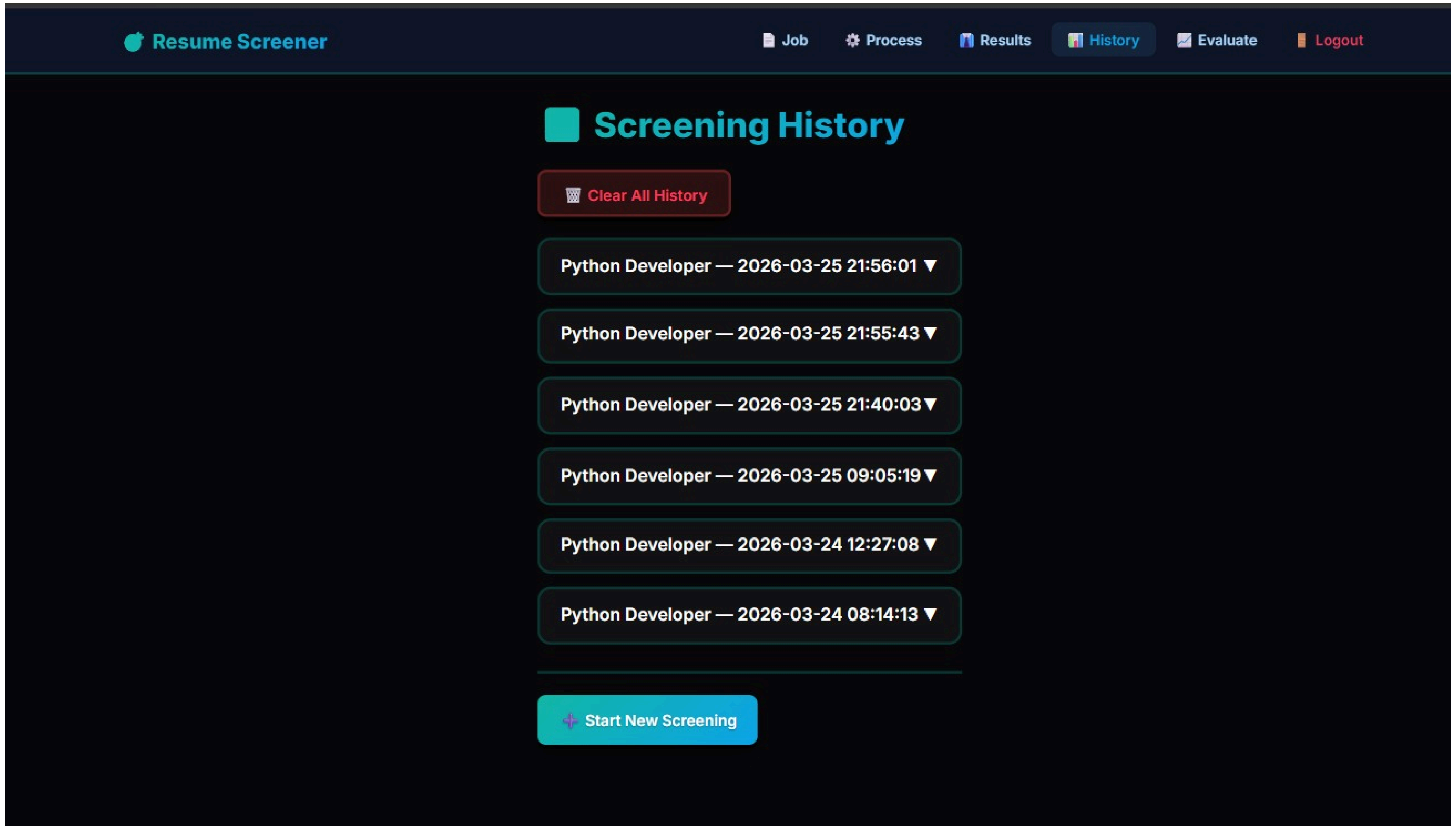
This page displays ranked and shortlisted candidates with match scores, allows threshold adjustment, comparison, and supports result export and notifications.



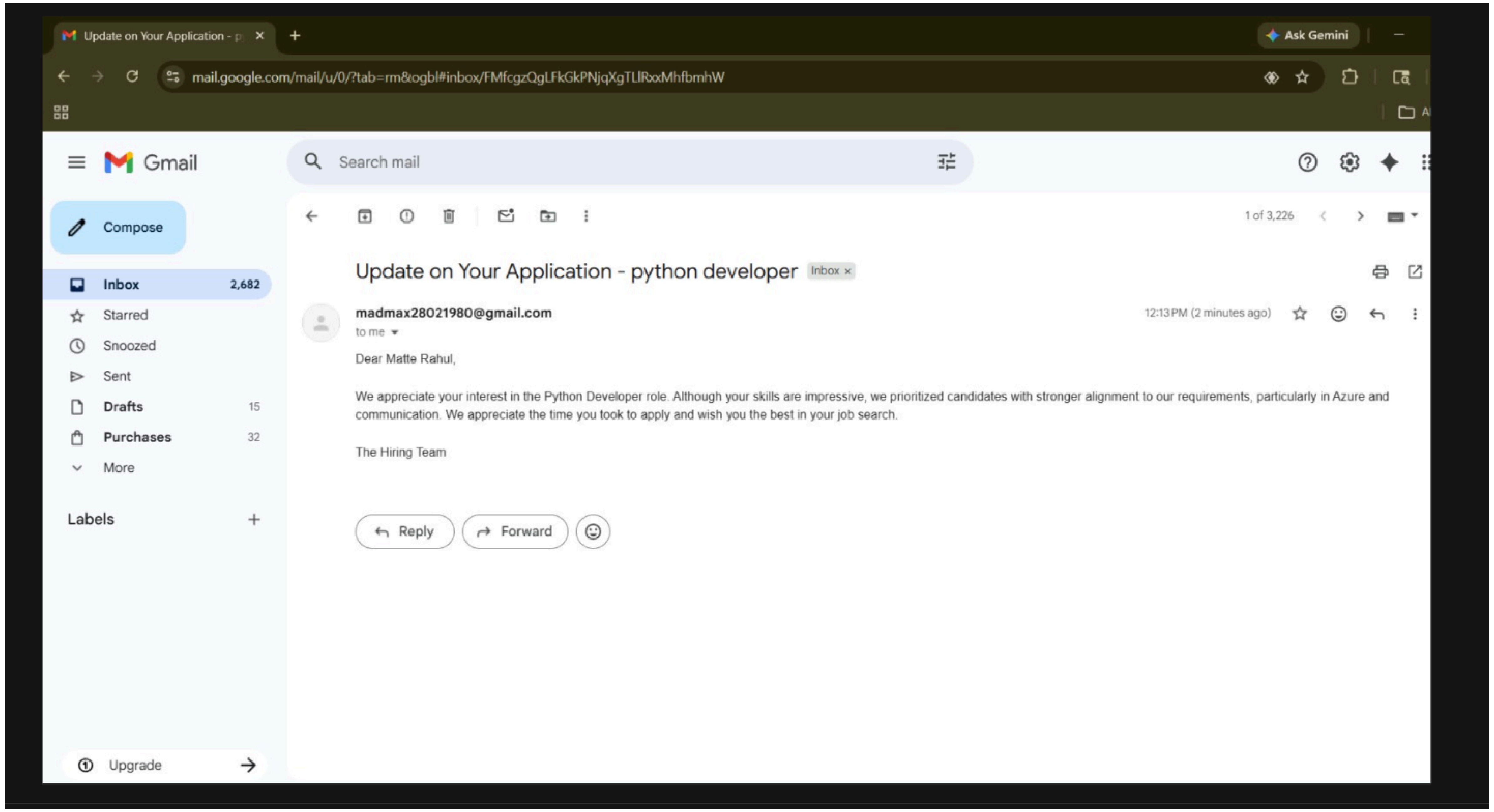
The XAI module explains candidate rankings by showing skill match, experience alignment, confidence level, and recommendations for transparent decision-making.



This page stores past screening sessions, allowing recruiters to review, manage, and track previous results efficiently.



The system automatically sends personalized emails to candidates with selection status and explanation, ensuring transparent and efficient communication.



System Overall Performance

To evaluate the effectiveness of the proposed system, a qualitative comparison was made between traditional resumescreening methods and the AI-based screening approach.

Table 10.1: Performance Improvement of the Proposed Resume Screening System

Metric	**Traditional Screening**	**AI-Based Resume Screening**
Screening Speed	Manual and time-consuming	Fast automated processing
Accuracy	Limited explanation	Explainable AI-based insights
Transparency	Manual planning requires more time	Automated AI scheduling improves efficiency
Scalability	Limited handling capacity	Handles large volumes efficiently

Conclusion

In this paper, we present an intelligent resume screening system that applies AI techniques to improve the recruitment process. Unlike traditional keyword-based methods, the proposed system uses semantic understanding through LLaMA to match resumes with job descriptions more accurately. By generating embeddings and applying similarity-based ranking, the system effectively identifies the most relevant candidates. The integration of Explainable AI ensures transparency by providing clear justifications for each decision. Overall, the system enhances efficiency, reduces manual effort and bias, and supports fair and reliable hiring. It contributes towards building a more accurate, transparent, and scalable recruitment process.

THANK YOU